

TrES-3b

R-0517

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240823 · created 2026-07-20T22:32:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460545.7349 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.056
Transit depth (Rp/Rs)^2	3.78 +/- 2.16 [%]
Orbital Inclination (inc)	80.58 +/- 2.62
Airmass coefficient 1 (a1)	0.963 +/- 0.033
Airmass coefficient 2 (a2)	0.025 +/- 0.035
Scatter in the residuals of the lightcurve fit is	3.39 %
Best Comparison Star	#3 - [380, 445]
Optimal Method	PSF photometry
Transit Duration (day)	0.039 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.194 ± 0.056 vs 0.16309 (deviation 0.55σ)
Duration fit vs published	0.039 d vs 0.0567 d (deviation 0.63σ)
Mid-time O–C	3.1 min
Depth SNR	3.5
Residual scatter	3.39 %

Light curve

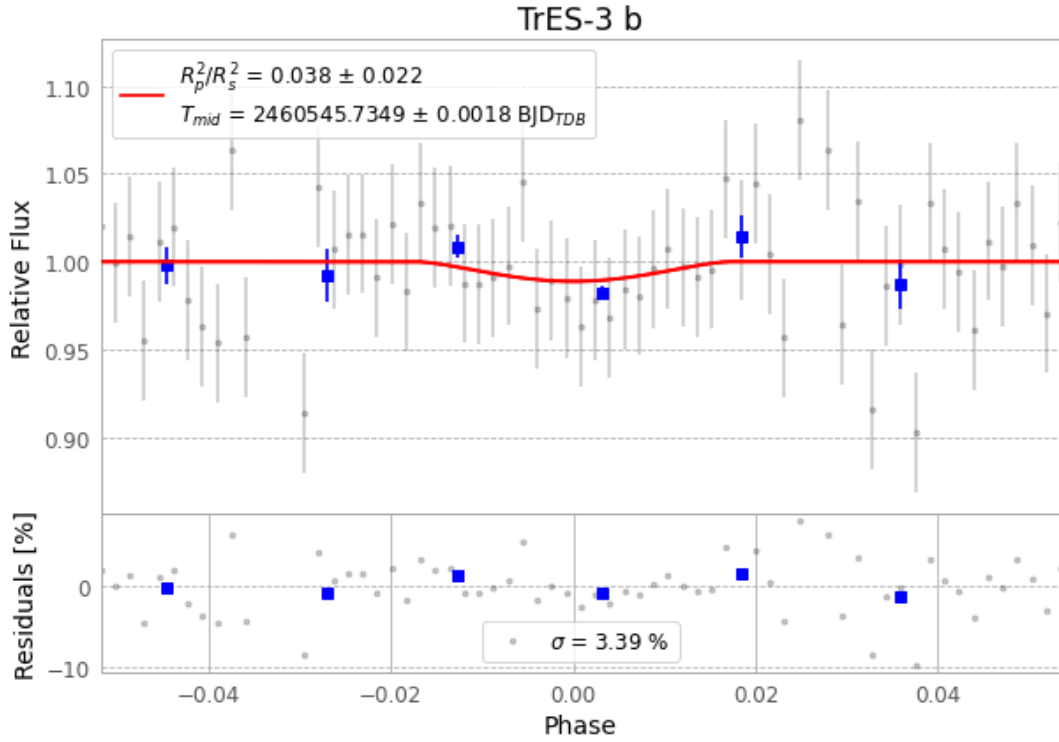


Figure 1 — FinalLightCurve_TrES-3 b_2024-08-22.png (SHA-256 ab841dd3932257ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 82329cc4afbe54c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

67 FITS frames (44 MB), TRES-3240823035715.FITS → TRES-3240823071515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240823.log (4d17254da28a...), FinalLightCurve_TrES-3 b_2024-08-22.png (ab841dd39322...), FinalLightCurve_TrES-3 b_2024-08-22.csv (28037ee3c4b0...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 22:32Z.